

Q1: 16 March (Shift 1) - Numerical

The decomposition of formic acid on gold surface follows first order kinetics. If the rate constant at 300 K is $1.0 \times 10^{-3} \text{ s}^{-1}$ and the activation energy $E_a = 11.488 \text{ kJ mol}^{-1}$, the rate constant at 200 K is $\underline{\hspace{1cm}} \times 10^{-5} \text{ s}^{-1}$. (Round off to the Nearest Integer).

(Given : $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)

Q2: 16 March (Shift 2) - Numerical

A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is $\underline{\hspace{1cm}}$ min. (Round off to the Nearest Integer).

Q3: 17 March (Shift 1) - Numerical

For a certain first order reaction 32% of the reactant is left after 570 s. The rate constant of this reaction is $\underline{\hspace{1cm}} \times 10^{-3} \text{ s}^{-1}$. (Round off to the Nearest Integer).

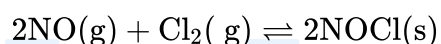
[Given : $\log_{10} 2 = 0.301, \ln 10 = 2.303$]

Q4: 17 March (Shift 2) - Numerical

The reaction $2 \text{ A} + \text{B}_2 \rightarrow 2 \text{ AB}$ is an elementary reaction.

For a certain quantity of reactants, if the volume

of the reaction vessel is reduced by a factor of 3, the rate of the reaction increases by a factor of $\underline{\hspace{1cm}}$. (Round off to the Nearest Integer).

Q5: 18 March (Shift 1) - Numerical

This reaction was studied at -10°C and the

following data was obtained

Questions with Answer Keys

MathonGo

run	$[\text{NO}]_0$	$[\text{Cl}_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[\text{NO}]_0$ and $[\text{Cl}_2]_0$ are the initial concentrations and r_0 is the initial reaction rate. The overall order of the reaction is _____.

(Round off to the Nearest Integer).

Q6: 18 March (Shift 2) - Numerical

A reaction has a half life of 1 min. The time

required for 99.9% completion of the reaction is _____ min. (Round off to the Nearest integer)

[Use : $\ln 2 = 0.69$, $\ln 10 = 2.3$]

Answer Key**Q1 (10)****Q2 (108)****Q3 (2)****Q4 (27)****Q5 (3)****Q6 (10)**

Hints and Solutions

MathonGo

Q1 (10)

$$K_{300} = 10^{-4} \quad K_{200} = ?$$

$$E_a = 11.488 \text{ kJ/mole} \quad R = 8.314 \text{ J/mole} \cdot \text{K}$$

$$\text{so } \ln \left(\frac{K_{300}}{K_{200}} \right) = \frac{E_a}{R} \left(\frac{1}{200} - \frac{1}{300} \right)$$

$$\text{so } \frac{K_{300}}{K_{200}} = 10$$

$$\ln \left(\frac{K_{300}}{K_{200}} \right) = \frac{11.488 \times 1000 \times 100}{8.314 \times 200 \times 300}$$

$$= 2.303$$

$$= \ln 10$$

$$K_{200} = \frac{1}{10} \times K_{300} = 10^{-4}$$

$$= 10 \times 10^{-5} \text{ sec}^{-1}$$

Q2 (108)

$$\text{Given } t_2 = 54 \text{ min}$$

$$T_{1/2} = 18 \text{ min}$$

A

B

$$t = 0 \text{ 'x' M}$$

$$t = 0 \text{ 'x' M}$$

$$\Rightarrow \text{To calculate : } [A_t] = 16 \times [B_t] \dots (1) \text{ time} = ?$$

$$\Rightarrow \text{For I order kinetic : } [A_t] = \frac{A_0}{(2)^n}$$

 $n \rightarrow$ no of Half lives

$$\Rightarrow \text{Now from the relation (1)}$$

$$[A_t] = 16 \times [B_t]$$

$$\Rightarrow \frac{x}{(2)^{n_1}} = \frac{x}{(2)^{n_2}} \times 16 \Rightarrow (2)^{n_2} = (2)^{n_1} \times (2)^4$$

$$\Rightarrow n_2 = n_1 + 4 \Rightarrow \frac{t}{(t_{1/2})_2} = \frac{t}{(t_{1/2})_1} + 4$$

$$\Rightarrow t \left(\frac{1}{18} - \frac{1}{54} \right) = 4 \Rightarrow t = \frac{4 \times 18 \times 54}{36}$$

$$\Rightarrow \boxed{t = 108 \text{ min}}$$

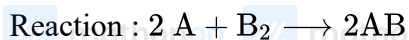
Q3 (2)

For 1st order reaction,

$$K = \frac{2.303}{t} \cdot \log \frac{[A_0]}{[A_t]} = \frac{2.303}{570 \text{ sec}} \cdot \log \left(\frac{100}{32} \right)$$

$$= 1.999 \times 10^{-3} \text{ sec}^{-1} \approx 2 \times 10^{-3} \text{ sec}^{-1}$$

Q4 (27)



As the reaction is elementary, the rate of reaction is $r = K \cdot [\text{A}]^2 [\text{B}_2]$

on reducing the volume by a factor of 3, the concentrations of A and B₂ will become 3 times and hence, the

rate becomes $3^2 \times 3 = 27$ times

of initial rate.

Q5 (3)

$$r = k[\text{NO}]^m[\text{Cl}_2]^n$$

$$= k(0.1)^m(0.1)^n$$

$$= k(0.1)^m(0.2)^n \dots (1)$$

$$= k(0.2)^m(0.2)^n \dots (3)$$

$$n = 1$$

$$m = 2$$

$$m + n = 3$$

Q6 (10)

$$\frac{t_{99.9\%}}{t_{50\%}} = \frac{\frac{1}{K} \ln \frac{100}{0.1}}{\frac{1}{K} \ln 2}$$

$$= \frac{\ln 1000}{\ln 2} \times t_{50\%}$$

$$= \frac{3 \ln 10}{\ln 2} \times 1$$

$$= \frac{3 \times 2.3}{0.69} = 10$$